

Dane Z. Dalton

Location: Huntsville, AL

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**Willing to be sponsored for TS/SCI CI of FS polygraph clearance*

<https://www.danezdalton.com/>

Work Experience

Adventures with Mr. Math, Remote - *Software Developer (Jan 2023 - Present)*

- Build a multi-faceted E-Learning Portal for over one thousand students, parents, TA's, and teachers of Mr. Math. Full life cycle development with ReactJS and TailwindCS, Dockerified for production.
- Functionality: Sign in, Sign up, Auth, Class Scheduling/Rescheduling, customized Interactive Worksheets, and a Mailbox System.
- Designed a friendly UI/UX for a primarily Elementary/Middle Schooler user-base. Important factors are making it easy to use, and cover as many edge cases as possible.
- Engineered a client-side caching system for the call-heavy worksheets. This improves the fluidity of working between multiple worksheet problems at the same time, and reduces stress on the server.
- Organize and streamline the dev experience with 3rd party tools including: Prettier, MaterialUI, date-fns, react-icons, and Axios. Utilizing these tools helps me quickly and efficiently build new features.
- Create and Lead a QA Process. Collaboration with a Lead QA Engineer helped me mold this process to our needs.
- Debug both frontend and our .NET backend when issues arise while making RESTful API calls. By creating better logs I give better feedback to our backend developer that leads to quicker system fixes.

Adventures with Mr. Math, Remote - *Director of Teaching Assistants (July 2022 - Present)*

- Ensure smooth operation for all of Mr. Math's TA needs. Organize 60+ TA's and their schedules so that teachers and admins can focus on the students and national contests.
- Scheduling, management, administration, organization, and interviewing skill growth.
- Create a C# program to calculate the TA hours at the end of each month using the Google Sheets API. This saves me 4-6 hours per month, allowing more time to spend working on the Portal. I save the Not-For-Profit money by reducing the odds of mispayment, and \$7,000 - \$11,000 per year for the labor of running hours.

Projects & Extracurriculars

My Portfolio, <https://www.danezdalton.com/> : A small portfolio to showcase my passion for creativity, and some of the projects I've worked on. Since using this portfolio to push my UI design limits, I've become much faster at solving UI/UX problems at work, allowing me more time to focus on other issues like security, pushing to production, and optimizing my logic.

Custom LED Light String: Decorative LED strip wired to an Arduino, using Arduino code (pseudo C++) and RGB values to create different light patterns. A semester-long project class, entirely self-directed and self-taught.

Etch-A-Sketch: A web app written in JS that provides an interactive color palette with an opacity setting that allows for modification of a sketch grid on a 'mouseover' event. The user-friendly UI/UX led to the opportunity to create the Mr. Math Portal frontend.

Knight Travails: An application written in Ruby to find the fastest path for a knight in a chess game to move from one square to another. Building a graph breadth first, the first occurrence of a finished game is found and the parent nodes are traced to output the quickest path from square A to square B.

Rock Climbing: I route-set for nearly a year at my local climbing gym. Creating challenges that other climbers get to explore and experience is incredibly rewarding. What I love about climbing is the opportunity to spend all day outdoors, set goals, study every minutia to get the most out of each rock hold, and perfecting my movement to top the climb.

Education

Iowa State University, Ames, IA | 2021 | Bachelor of Science, Environmental Science | Minor in Biology

Skills

Software Skills: ReactJS, TailwindCSS, JavaScript, TypeScript, HTML, CSS, UI/UX, Docker, Git, Vite, C#, SQL, SSMS, Azure, QA Lead, Ruby, Rails, Linux OS, REST API, Postgres, Bootstrap, C, C++, CAD, GIS, LaTeX, Salesforce Certified Administrator, Excel, Google Workspace, Godot, GDScript

Hardware Skills: Arduino, Logic Gates, Solder, Breadboard, PC Building, Multimeter, Oscilloscope, Isometric Sketching, Autoclave